NETWORK MANAGEMENT SYSTEM

# Statement of Work

Central monitoring and switch management across the agreed campus network

| Client | Supplier | Delivery estimate |
| --- | --- | --- |
| [Legal name] | [Legal name] | About 12 weeks |

This work is to deliver a usable first release of the NMS for the client's agreed network. Operators will use one web application to see devices and links across the agreed campuses, follow alarms and performance, and prepare or make approved switch changes. The supplied switch manuals will be used to build and qualify the initial device profiles.

The client has asked for coverage of three switch types. The references include ATS command material, a second switch documentation bundle, and the Routing Switch CLI reference V3.3.1. At kickoff, both teams will confirm the exact manufacturer, model and firmware represented by each type. The manuals are technical references; they do not, by themselves, make every command a supported NMS feature.

## Included in the NMS release

- A single web console with user login, role-based access for the agreed operator roles, and separate views for each campus and the overall network.
- A central device register with address, site, asset tag, model, serial number, firmware, CPU and memory fields where the device exposes them. Values will show their source and last-seen time. Changes made in the NMS will be attributable to the signed-in user.
- Automatic operating-system and software-family identification during onboarding. The NMS will use available evidence such as SNMP system description, reported vendor/model and an approved read-only CLI query. It will show the evidence and time observed, select a matching profile only when the match is clear, and ask an operator to review unknown or conflicting results.
- A maintained device capability matrix showing which models, OS releases, telemetry fields and configuration operations have been tested. Unsupported items will be shown as such in the interface.

## Deployment basis

The delivery covers one agreed NMS environment and the first release for the device and campus counts recorded at Milestone 1. The client will provide the server or virtual machine, network access and approved operating environment. RHEL compatibility and sizing will be confirmed against the client's exact version and runtime; this SOW does not claim certification before that validation.

Use of NMS as a custom-built product assumes written client approval if procurement requires an OEM or established commercial NMS. This SOW does not establish that product eligibility.

# What the system will do

## Monitoring and device inventory

- Onboard approved devices from the provided inventory and collect availability and health using the agreed ICMP and SNMP methods. SNMPv3 will be used where the client supplies and enables it on supported devices; any protocol or model limitation will be recorded.
- Collect interface state, traffic counters, errors and capacity information. Show receive/transmit rates and utilization where valid counters and interface speed are available. Collect CPU, memory, temperature or other vendor-specific values only where a tested MIB, API or CLI source exists.
- Keep time-series history for the agreed retention period and show current and historical device and interface performance. Record the source and freshness of values so missing data is not mistaken for a healthy reading.
- Allow approved inventory imports or additions, site assignment, search and filtering. Subnet discovery, if required, will use ranges and scan limits approved during design.

## Topology and campus views

- Build link relationships from LLDP or another agreed device source and present them in campus and network views. Show link state and available traffic/utilization data, with a clear indication when topology or telemetry is stale.
- Allow operators to mark important devices or links and open device, interface and alarm details from the topology. Endpoint or access-terminal visibility is limited to information the switches expose and the client authorizes the NMS to collect.

## Alarms and event handling

- Provide configurable thresholds for the agreed health and performance measures, with Critical, Major, Minor and Warning severity levels, plus duration and clear conditions to reduce short-lived noise.
- Collect supported polling alarms and, where access is provided, SNMP traps and syslog. Show active and historical events with device/site context, time, severity, acknowledgement and resolution state.
- Provide email and webhook notifications for the agreed event rules, including delivery status and retry behavior. Any action that changes a device or calls an external script will require a named, separately approved workflow.

## Reports

- Include configurable inventory, device-health, interface-usage and alarm reports with filters for campus, device and time period. Operators can choose the agreed columns and export reports to CSV or PDF.
- Provide event and device history for observed changes and NMS actions, including user, timestamp and before/after values where the NMS has both values.

# Switch configuration and change records

The configuration area will use the confirmed capability matrix rather than expose commands indiscriminately. It will cover the operations selected by the client from the supplied references and validated against representative devices.

- For each approved profile, provide structured configuration for the agreed initial areas: basic and management settings, interfaces, VLAN membership, link aggregation, spanning tree, QoS and selected security or network features. The exact commands and fields for each model/firmware will be listed in the Milestone 1 matrix.
- Check required values and profile-specific limits before a change can proceed. Show the target device, detected OS/profile, proposed change and generated device-specific commands to the operator.
- Support a review and approval step before applying a change. Where a multi-device change is approved, show the target list and per-device result; do not report a partial rollout as successful. The agreed security operations may include ACLs, 802.1X, port security, DHCP snooping, QoS, and connection restriction or quarantine actions when the device supports them.
- For operations enabled for live application, capture the existing configuration or relevant state before the change, record the result and verify the expected state afterward. Offer rollback only where the device and operation have a tested recovery method. If safe rollback is unavailable, say so before approval.
- Keep a configuration history with device, user, time, operation, result and the available before/after difference. Protect credentials and configuration archives according to the agreed security design.
- Include firmware version and change history. Firmware image distribution or upgrades are limited to the models, image sources and recovery steps specifically approved and tested in the capability matrix; no unverified image will be installed.

## Traffic analysis

If the client’s switches can export NetFlow, IPFIX or sFlow, the NMS will receive and report on one agreed flow format for the accepted device matrix. The client will identify the exporter format, volume and retention requirement during discovery. Traffic reports will cover agreed top devices, protocols and conversations. Flow analysis is not inferred from SNMP interface counters.

## Delivery and handover

- Install and configure the agreed release in one client-provided environment; supply deployment settings and a short operating guide.
- Run the agreed acceptance scenarios with representative devices from the confirmed matrix and demonstrate monitoring, OS identification, topology, alarms, reports and the approved configuration path.
- Hand over the supported-device matrix, known limitations, configuration notes, operator guide and a release record. Provide one operator handover session.

Work outside the agreed device matrix, integration list or operation list will be estimated and approved in writing before work starts. This keeps the initial delivery tied to tested behavior instead of an open-ended promise to support every command in every manual.

# Milestones and acceptance

The current schedule is approximately 12 weeks from kickoff. Milestone 1 will confirm the device count, access, test plan and acceptance checklist; the schedule will be updated if those inputs materially change.

| MILESTONE | DELIVERABLES FOR REVIEW | TARGET |
|---|---|---|
| **1. Scope and design** | Confirmed device/model/firmware and campus matrix; OS identification approach; monitoring sources and retention; configuration and security operations; access plan; acceptance scenarios. | Weeks 1-2 |
| **2. Build** | Review release covering inventory, OS identification, monitoring, topology, alarms, reports, flow analysis and the configuration workflows recorded in the approved capability matrix. | Weeks 3-9 |
| **3. Validation and handover** | Demonstration against the accepted scenarios; correction of material defects; deployment to the agreed environment; operating guide, supported-device matrix and handover session. | Weeks 10-12 |

## Client inputs

The client will provide the device and firmware list, representative test access, approved credentials and network routes, the target server environment, notification endpoints, and a technical contact for profile questions. The client will arrange any maintenance windows and approvals needed for testing. Delays in these dependencies may move the schedule.

## Review and acceptance

The client will review each milestone within five business days and either accept it in writing or send one consolidated list of material gaps against the agreed deliverables. The supplier will address confirmed material gaps and resubmit. New devices, features, reports or integrations require written change approval and may affect the schedule.

## Scope boundaries

The supported models and operations are those recorded in the accepted capability matrix. Hardware, third-party licenses, switch support contracts, travel, high-availability engineering, formal security certification, 24x7 support, and integrations not listed in the matrix are outside this SOW. Live configuration and firmware actions are enabled only for device/operation combinations that pass the agreed test and approval process.

## Procurement condition

If the client specification requires an established commercial or OEM-backed NMS, the client must approve NMS in writing as an exception before kickoff. This statement does not itself establish OEM backing or commercial product eligibility.

| CLIENT | SUPPLIER |
|---|---|
| Name: _____<br>Title: _____<br>Signature: _____<br>Date: _____ | Name: _____<br>Title: _____<br>Signature: _____<br>Date: _____ |